Density Dependent Marked Point Process Framework (DenMark) Tutorial

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This tutorial shows how to use **DenMark** to quantify the correlation between gene expression and cell density. We also show how to identify the density-correlated genes (DCG) in the single-cell resolution spatial transcriptomics (ST) dataset.

We use the 10X Xenium breast cancer dataset as an example **(dataset can be found here)**. Section 1 describes data loading and ROI selection, followed by coordinate system normalization and data preprocessing in Section 2. Section 3 presents the quantification of the relationship between gene expression and cell density, while Section 4 focuses on the identification of DCGs.

We start with two files:

- A 2D spatial locations of cells worksheet (cell ID $\times$ 2D coordinate);
- A raw gene expression over the cells worksheet (cell ID $\times$ gene expression).

All the content in this tutorial is run under a PC with Window 11 operation.

Prepare packages and functions

DenMark is built on a stan environment, so a stan installation environment is needed. This include the installation of *stan*, *rstan*, and *cmdstanr*.

```
# [Code Block 0.1] Installation of rstan, and cmdstanr

#  ---- Dependent on the operation, the installation might be different ----
# Please refer to the installation page:  https://mc-stan.org/install

rm(list=c())
# Finally, you are supposed to load the packages:
library('rstan')
library('cmdstanr')
```

Some packages are also useful for the data analysis, make sure you have downloaded them before we get start.

```
# [Code Block 0.2] Load Other useful packages

# load other packages for visualization and DenMark analysis
```

```r
# (a) For data analysis:
library(reshape2)
library(dplyr)
library('MASS')
library(fields)
library(Matrix)
library(tidyverse)
library(magrittr)
library(tidybayes)
library(coda)
library(loo)

# (b) For visualization:
library('ggplot2')
library(cowplot)
library(viridis)
library(rethinking)
library(ggridges)
library(ggrepel)

# (c) Self-built posterior summary functions:
q2.5 <- function(x) quantile(x, prob = 0.025)
q25 <- function(x) quantile(x, prob = 0.25)
q50 <- function(x) quantile(x, prob = 0.50)
q75 <- function(x) quantile(x, prob = 0.75)
q97.5 <- function(x) quantile(x, prob = 0.975)

# (d) Matern32 kernel: covariance function for Matern (nu = 3/2)
matern32 <- function(d, sigma, rho) {
  sigma^2*(1 + sqrt(3) * d / rho) * exp(-sqrt(3) * d / rho)
}

# (e) cmdstanr settings for showing more rows
options(cmdstanr_max_rows = 20)

# Set the workplace
setwd('C:/Users/xumc7/OneDrive/Desktop/Tutorial')

# (f) DenMark functions and stan file directory:
source('DenMark/DenMark-functions.R')
denmark_stan       <- 'DenMark/DenMark-M2.stan'
denmark_simple_stan <- 'DenMark/DenMark-M1.stan'
```

1. Load single-cell resolution ST data, select hyperparameters

Now load the two matrices. We use the **In Situ Sample 1, Replicate 1** as an example.

```r
# [Code Block 1.1] Load two matrices

# (a) Cell ID * Gene expression:
cell_gene <- read.csv('cell_by_gene_expression.csv')
# (b) Cell ID * 2D spatial locations:
```

```r
cell_spatial <- read.csv('cell_spatial.csv')

# check the dimensions:
dim(cell_gene) # cell counts * gene numbers
```

```
## [1] 167780    322
```

```r
# check the gene names:
colnames(cell_gene)[2:6] # the first 5 genes in the dataset
```

```
## [1] "ABCC11" "ACTA2"  "ACTG2"  "ADAM9"  "ADGRE5"
```

Then hyper parameters are user-defined: the grid number controls the grid approach approximation precision, the number of basis functions at the x-/y- axis and the boundary choice

```r
# [Code Block 1.2] Select hyper-parameters

# Hyperparameters that influence the approximate precision:
grid_size<- 15      # grid cells at the x-/y- direction
basis_choice <- 25  # number of basis functions for the HSGP. higher number makes a higher HSGP approxi
boundary_choice <- 1.5 # range from [1, Inf); boundary factor for HSGP to avoid boundary effect
```

2. ROI Selection, Coordinate Normalization and Data Preprocessing

The ROI is chosen as the research interest needs. For the HSGP to be well-defined, we need to move the ROI under the original coordinates to an ROI symmetric about the origin. In this section we describe running DenMark over a global ROI (the whole tissue) for the large range analysis.

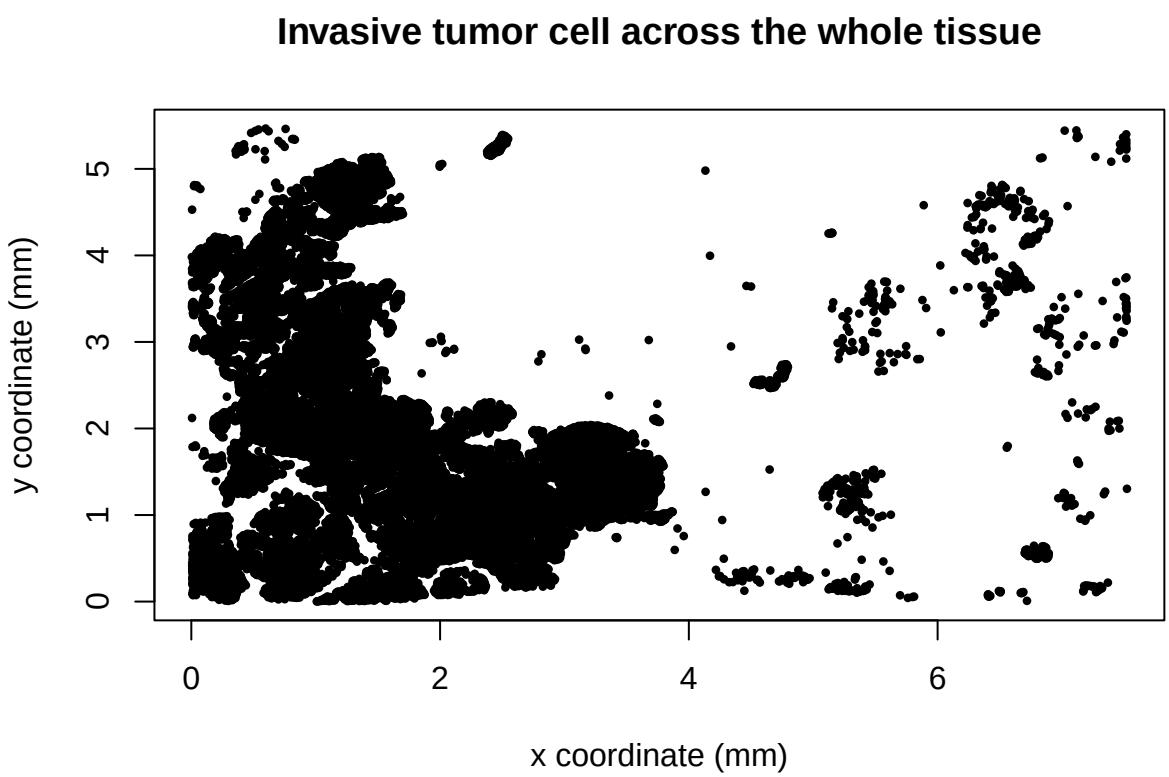
```r
#  [Code Block 2.1] Scale the coordinates into mm*mm measurement
cell_spatial$x<- cell_spatial$x/1000
cell_spatial$y<- cell_spatial$y/1000

# check the dataset:
head(cell_spatial)
```

```
##   X         x         y       celltype
## 1 0 0.8462731 0.3275442         DCIS_2
## 2 1 0.8253991 0.3295058         DCIS_2
## 3 2 0.8484960 0.3314183      Unlabeled
## 4 3 0.8242057 0.3321048 Invasive_Tumor
## 5 4 0.8431182 0.3315164         DCIS_2
## 6 5 0.8486596 0.3347365      Unlabeled
```

2b Large range analysis by a global ROI

We select the invasive tumor cell as the example.

```r
# [Code Block 2b.1] Extract the invasive tumor and visualize

# extract the cell type:
center_slice_cell_location_tumor <- cell_spatial[which(cell_spatial$celltype=='Invasive_Tumor'),]

# extract the corresponding genes at the same locations
gene_tumor <- cell_gene[which(cell_spatial$celltype=='Invasive_Tumor'),]

# visualize the cell locations: each black dot is a cell location
plot(center_slice_cell_location_tumor$x,
     center_slice_cell_location_tumor$y,
     main = 'Invasive tumor cell across the whole tissue',
     xlab='x coordinate (mm)',
     ylab='y coordinate (mm)',
     pch = 16,
     cex = 0.6)
```




```r
# [Code Block 2b.2] Quality control
# filter out the genes with (1) the lowest 5% , (2) zeros with more than 5% locations
gene_tumor <- gene_tumor[,-1]

# Convert to matrix (if not already)
mat <- as.matrix(gene_tumor)

# (1) Compute variance per gene
```

```r
gene_var <- apply(mat, 2, var)

# Keep top 95% high-variance genes
var_threshold <- quantile(gene_var, 0.05)
genes_high_var <- names(gene_var[gene_var >= var_threshold])

# (2) Non-zero filter (>=5%)
nonzero_fraction <- colSums(mat > 0) / nrow(mat)
genes_nonzero <- names(nonzero_fraction[nonzero_fraction >= 0.05])

# Combine both filters
genes_keep <- intersect(genes_high_var, genes_nonzero)

# Filtered dataset
gene_tumor_filtered <- mat[, genes_keep, drop = FALSE]

# Check the number of filtered genes
# length(colnames(gene_tumor_filtered)) # 174 filtered genes

# check first 10 filtered genes
colnames(gene_tumor_filtered)[1:10]
```

```
##  [1] "ABCC11"   "ACTA2"    "ACTG2"    "ADAM9"    "AGR3"     "AIF1"
##  [7] "ANKRD28"  "ANKRD29"  "ANKRD30A" "APOBEC3B"
```

The combined filtered dataset is defined.

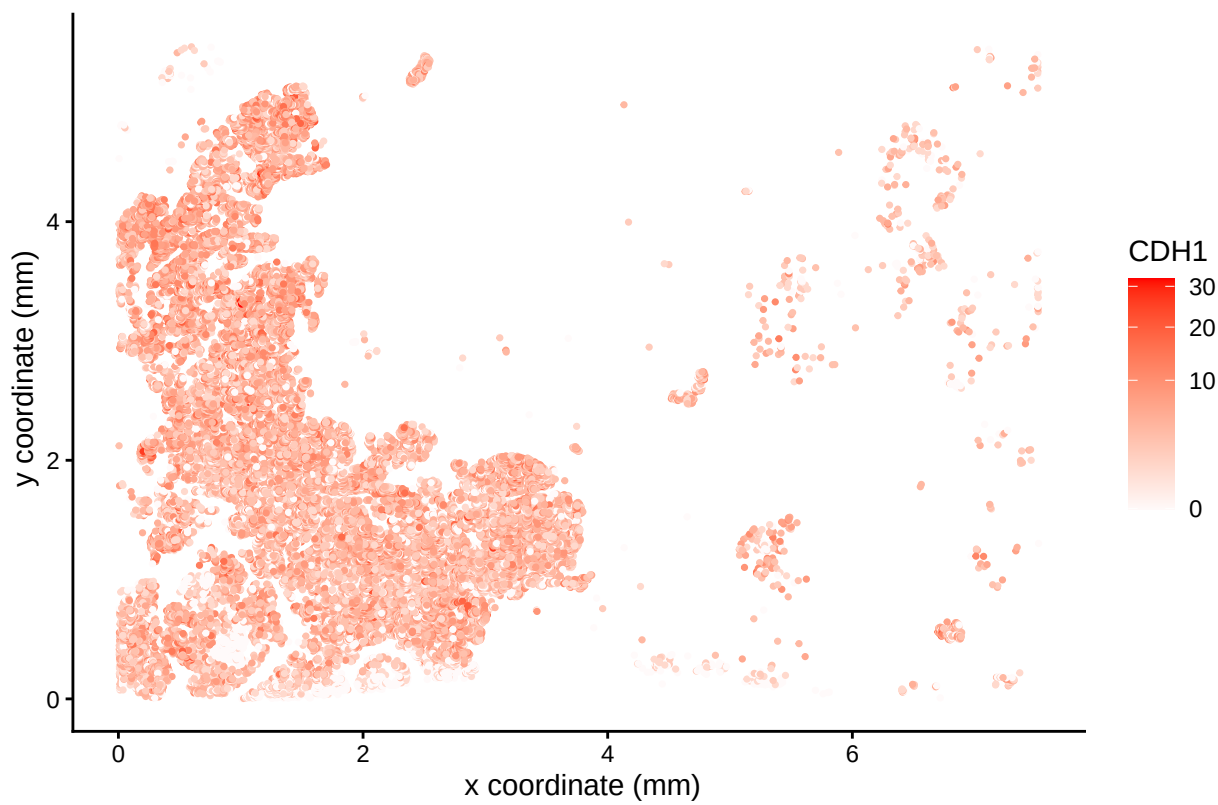
```r
# [Code Block 2b.3] Rearrange the dataset
final_short <- as.data.frame(cbind(center_slice_cell_location_tumor$x, # x-coordinate
                                   center_slice_cell_location_tumor$y, # y-coordinate
                                   gene_tumor_filtered # filtered gene expression
                                   ))

# rename the first two columns
colnames(final_short)[1:2]<- c('x','y')

# check the first few columns
head(final_short[,c(1:5)])
```

```
##            x         y ABCC11 ACTA2 ACTG2
## 4  0.8242057 0.3321048      0     0     0
## 19 0.8404538 0.3750625      0     1     0
## 20 0.8342422 0.3784625      0     0     0
## 22 0.8352884 0.3846740      0     0     3
## 23 0.8241894 0.4106644      0     0     0
## 35 0.8353702 0.4604058      0     0     0
```

Here, we are going to show the analysis for one typical gene expression, say, gene $CDH1$.

```r
# [Code Block 2b.4] Select your typical gene for the analysis

# Please refer to the selected informative gene list for selecting your gene of interest:
```

```r
#  colnames(final_short)[-c(1,2)]

gene_choice <- 'CDH1'
df_short_select <- data.frame(final_short$x,
                              final_short$y,
                              final_short$CDH1)

colnames(df_short_select) <- c('x', 'y', gene_choice)


# we can visualize the gene expression pattern at the point pattern over the region:
ggplot(df_short_select, aes(x = x, y = y, color = CDH1)) +
  geom_point(size = 0.6, alpha = 1) +
  scale_color_gradient(low = "snow1", high = "red", trans = "sqrt") +
  labs(title="", x=" x coordinate (mm)", y=" y coordinate (mm)")+
  theme(text = element_text(size=20),
        strip.text.x = element_text(size=20),
        strip.text.y = element_text(size=20))+
  theme(legend.text = element_text(size=50))+
  theme_classic()
```




```r
# [Code Block 2b.5] Normalization of the locations

# the ROI is defined as [-L1, L1]*[-L2, L2]. The half range L1 and L2 are transformed from the original
L1<- max(abs(center_slice_cell_location_tumor$x-
```

```
                (min(center_slice_cell_location_tumor$x)+
                   max(center_slice_cell_location_tumor$x))/2 ))

L2<- max(abs(center_slice_cell_location_tumor$y-
                (min(center_slice_cell_location_tumor$y)+
                   max(center_slice_cell_location_tumor$y))/2 ))

# prepare the range parameter:
xRangeDat <- c(-L1,L1)
yRangeDat <- c(-L2,L2)
```

We then prepare the other input elements before running DenMark:

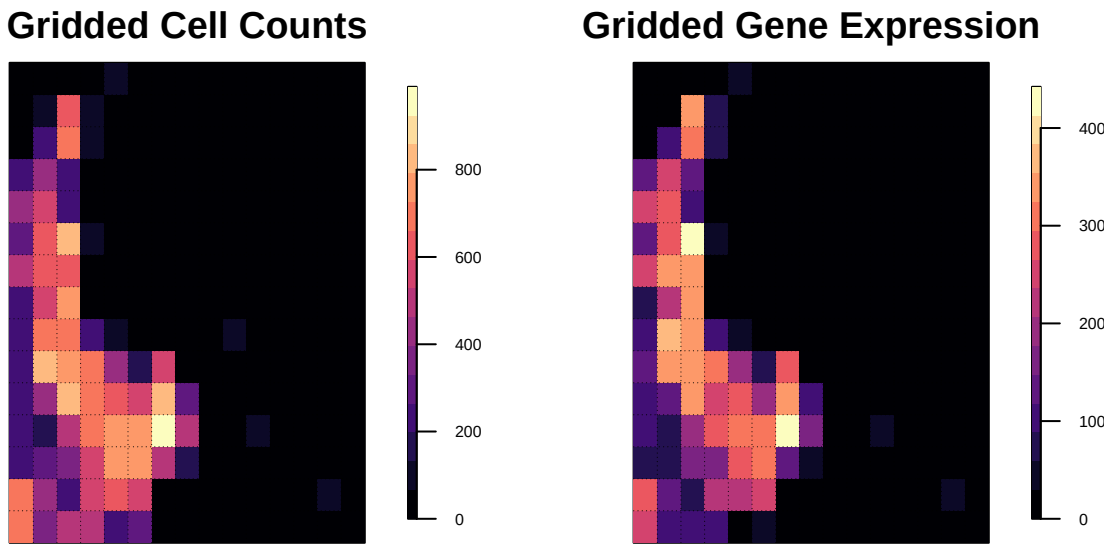
```
# [Code Block 2b.5] Calculate the input of DenMark

# Use functions 'buildgridpp' and 'grid_pts_centroid' from DenMark-function.R
buildgrid_result <- buildgridpp(grid_size = grid_size,
                                dataset = df_short_select)

# Note:
# buildgrid_result[[1]]: cell counts + gene expression counts at the 100 grids
# buildgrid_result[[2]]: matrix form for the cell counts
# buildgrid_result[[3]]: matrix form for the gene expression counts



grid_centroid_result <- grid_pts_centroid(grid_size = grid_size,
                                          dataset = df_short_select)

# Note:
# grid_centroid_result[[1]]: locations of the grids
# grid_centroid_result[[2]]: grid area
# grid_centroid_result[[3]]: the total number of grids
```

3. Univarite Gene Analysis: Quantify the density-expression association

Now we will apply the grid approach to analyze the discretized cell locations and gene expression. First, let's visualize those grid-level cell counts and gene expression.

```
# [Code Block 3.5] Check the grid visualization for the discretized data

# load the gridded cell counts and gene expression
loc_mat <- buildgrid_result[[2]]
marks_mat<- buildgrid_result[[3]]

# Helper to make breaks
make_breaks <- function(mat, n_colors = grid_size) {
  seq(min(mat, na.rm = TRUE),
      max(mat, na.rm = TRUE),
      length.out = n_colors + 1)
}
```

```r
#  Cell Counts | Gene expression
layout(matrix(1:2, nrow = 1), widths = c(1, 1), respect = F)

# Points
par(mar = c(8, 4, 2, 3))
image.plot(
  loc_mat,
  breaks = make_breaks(loc_mat, grid_size),
  col = viridis(grid_size, option= 'magma'),
  main = "Gridded Cell Counts",
  xlab = "",
  ylab = "",
  useRaster = FALSE,
  legend.width = 0.6,
  legend.mar = 2,
  xaxt = "n",
  yaxt = "n",
  axis.args = list(cex.axis = 0.5)
)
grid(nx = ncol(loc_mat), ny = nrow(loc_mat), col = "black", lwd = 0.5)

# Marks
par(mar = c(8, 4, 2, 3))
image.plot(
  marks_mat,
  breaks = make_breaks(marks_mat, grid_size),
  col = viridis(grid_size, option= 'magma'),
  main = "Gridded Gene Expression",
  xlab = "",
  ylab = '',
  useRaster = FALSE,
  legend.width = 0.6,
  legend.mar = 2,
  xaxt = "n",
  yaxt = "n",
  axis.args = list(cex.axis = 0.5)
)
grid(nx = ncol(marks_mat), ny = nrow(marks_mat), col = "black", lwd = 0.5)
```




```
# Reset the figure setting
par(mfrow = c(1, 1))
```

Next, we apply DenMark main function to the prepared data.

```
# [Code Block 3.5] Run DenMark function

# Recall:
# grid_size: the number of grids per x/y- axis
# dataset: containing (x, y) coordinates and expression at the selected ROI
# basis: the number of basis functions for HSGP per x/y- axis
# boundfactor: (1, Inf) boundary factor to get rid of boundary effect in HSGP
# L1: half range in the x-axis
# L2: half range in the y-axis
# denmark_version: Choose from
#   (1) complete version DenMark (density-expression correlated) stan file, or
#   (2) simplified DenMark version (density-expression independent) stan file


# Run the full DenMark version:
denmark_output_full <- DenMarkmain(grid_size = grid_size,
                                   dataset = df_short_select,
                                   basis = basis_choice,
                                   boundfactor = boundary_choice,
                                   L1 = L1,
```

```r
                              L2 = L2,
                              denmark_version = denmark_stan)
```

```
## Running MCMC with 4 parallel chains...
##
## Chain 1 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 2 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 3 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 4 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 1 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 3 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 4 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 2 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 1 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 3 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 4 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 2 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 1 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 3 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 4 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 2 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 1 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 3 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 4 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 2 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 1 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 4 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 3 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 2 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 1 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 1 finished in 884.7 seconds.
## Chain 4 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 4 finished in 887.5 seconds.
## Chain 3 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 2 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 3 finished in 888.2 seconds.
## Chain 2 finished in 888.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 887.2 seconds.
## Total execution time: 888.7 seconds.
```

```r
# As well as the simplified version (for model comparison usage):
denmark_output_simple <- DenMarkmain(grid_size = grid_size,
                              dataset = df_short_select,
                              basis = basis_choice,
                              boundfactor = boundary_choice,
                              L1 = L1,
```

```r
                                L2 = L2,
                                denmark_version = denmark_simple_stan)
```

```
## Running MCMC with 4 parallel chains...
##
## Chain 1 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 2 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 3 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 4 Iteration:    1 / 2000 [  0%]  (Warmup)

## Chain 2 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 1 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 4 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 3 Iteration:  400 / 2000 [ 20%]  (Warmup)
## Chain 2 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 1 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 4 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 3 Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 2 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 1 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 4 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 3 Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 2 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 1 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 4 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 3 Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 2 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 1 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 4 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 3 Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 2 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 2 finished in 774.9 seconds.
## Chain 1 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 1 finished in 776.3 seconds.
## Chain 4 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 4 finished in 785.6 seconds.
## Chain 3 Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 3 finished in 787.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 781.1 seconds.
## Total execution time: 787.8 seconds.
```

Now let's check the fitted results.

```r
# [Code Block 3.8] Check the posterior summary

# 1. check posterior summary for these parameters:
```

```r
# beta0, beta1: intercepts for cell density/ gene expression latent field
# ell[1], ell[2]: length of scale for cell density/ gene expression latent field
# Astar[2,1]: the density-expression correlation.
# sigma[1], sigma[2]: SD of the cell density and gene expression latent field
# rhat: criteria to check if convergence
# ess_bulk and ess_tail: effective sample sizes
fit_summary_full <- denmark_output_full$fit$summary(variables = c("beta0",
                                                                  "beta1",
                                                        "ell[1]", 'ell[2]',
                                                       'Astar[2,1]',
                                                       'sigma[1]','sigma[2]'),
                                        c("mean","sd",
                                          "q2.5","q25","q50","q75","q97.5",
                                          "rhat","ess_bulk","ess_tail"))

# transfer to the data frame
summary_full<- denmark_output_full$fit$draws(format = "df")



fit_summary_full
```

```
## # A tibble: 7 x 11
##   variable  mean     sd `2.5%` `25%` `50%` `75%` `97.5%`  rhat ess_bulk ess_tail
##   <chr>    <dbl>  <dbl>  <dbl> <dbl> <dbl> <dbl>   <dbl> <dbl>    <dbl>    <dbl>
## 1 beta0    3.81  0.711   2.39  3.36  3.82  4.26    5.21   1.00    1145.    1623.
## 2 beta1    0.860 0.120   0.609 0.786 0.866 0.941   1.08   1.00    1254.    1948.
## 3 ell[1]   0.756 0.0729  0.626 0.705 0.750 0.801   0.916  1.00    1110.    1564.
## 4 ell[2]   0.414 0.0769  0.283 0.359 0.407 0.461   0.577  1.00    1034.    1614.
## 5 Astar[2~ 0.523 0.106   0.286 0.457 0.533 0.597   0.708  1.00    1164.    1678.
## 6 sigma[1] 3.04  0.295   2.54  2.83  3.01  3.22    3.73   1.00     889.    1354.
## 7 sigma[2] 0.615 0.0655  0.504 0.571 0.611 0.654   0.753  1.00    1220.    2069.
```

```r
# 2. also check the posterior summary for the simplified DenMark version:
# beta0, beta1: intercepts for the cell density / gene expression latent field
# ell[1], ell[2]: length of scale for the cell density / gene expression latent field
# a11, a22: SD of the cell density and gene expression latent field
fit_summary_simple <- denmark_output_simple$fit$summary(variables = c("beta0",
                                                                      "beta1",
                                                        "ell[1]", 'ell[2]',
                                                       'a11', 'a22'),
                                        c("mean","sd",
                                          "q2.5","q25","q50","q75","q97.5",
                                          "rhat","ess_bulk","ess_tail"))
# transfer to the data frame
summary_simple<- denmark_output_simple$fit$draws(format = "df")


fit_summary_simple
```

```
## # A tibble: 6 x 11
##   variable  mean     sd `2.5%` `25%` `50%` `75%` `97.5%`  rhat ess_bulk ess_tail
```

```
##   <chr>    <dbl>  <dbl>  <dbl> <dbl> <dbl> <dbl>   <dbl> <dbl>    <dbl>    <dbl>
## 1 beta0    3.88  0.561   2.76  3.50  3.89  4.24    5.00   1.00    2893.    2747.
## 2 beta1    0.973 0.106   0.743 0.907 0.979 1.05    1.16   1.00    2238.    2022.
## 3 ell[1]   0.682 0.0541  0.584 0.645 0.680 0.718   0.792  1.00    1661.    2549.
## 4 ell[2]   0.515 0.0813  0.377 0.457 0.507 0.565   0.689  1.00    1338.    2296.
## 5 a11      2.60  0.181   2.28  2.48  2.59  2.72    2.98   1.00    1652.    2369.
## 6 a22      0.558 0.0573  0.462 0.519 0.553 0.590   0.684  1.00    1545.    2271.
```

```r
# [Code Block 3.9] Other fitting checks: WAIC and running time

# Show the WAIC:
print(paste0('The WAIC for the full DenMark model is: ',
             round(denmark_output_full$waic$waic, 2) ))
```

```
## [1] "The WAIC for the full DenMark model is: 2586.81"
```

```r
print(paste0('The WAIC for the simplified DenMark model is: ',
             round(denmark_output_simple$waic$waic, 2) ))
```

```
## [1] "The WAIC for the simplified DenMark model is: 2609.31"
```

```r
# Shoe the running time:
print(paste0('The running time for the full DenMark version is: ',
             round(denmark_output_full$time, 2),' mins.' ))
```

```
## [1] "The running time for the full DenMark version is: 14.81 mins."
```

```r
print(paste0('The running time for the simplified DenMark version is: ',
             round(denmark_output_simple$time,2), ' mins.' ))
```

```
## [1] "The running time for the simplified DenMark version is: 13.13 mins."
```

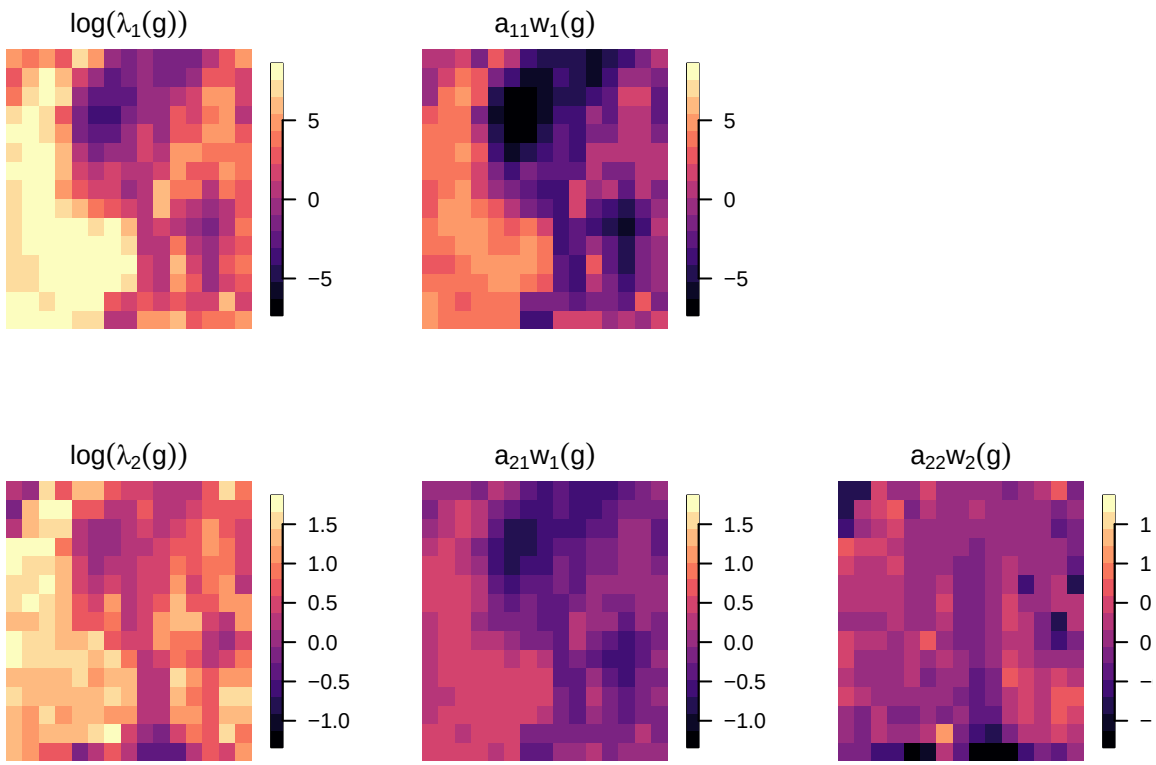
It is of interest to check the latent field decomposition for each grid. We visualize the decomposition for the fitted results under the full version DenMark.

```r
# [Code Block 3.10] Latent field decomposition

# summary of the posterior
samples_beta0<- summary_full[['beta0']]
samples_beta1<- summary_full[['beta1']]
samples_corr<- summary_full[['Astar[2,1]']]
samples_rho1<- summary_full[['ell[1]']]
samples_rho2<- summary_full[['ell[2]']]
samples_sigma1<- summary_full[['sigma[1]']]
samples_sigma2<- summary_full[['sigma[2]']]
# LMC matrix
samples_a11<- samples_sigma1
samples_a21<- samples_sigma2*samples_corr
samples_a22<- samples_sigma2*sqrt(1-samples_corr^2)
```

```r
# a11*w1= sigma1*w1
cal_a11w1<- function(i){
  w1<-denmark_output_full$fit$draws(variables = paste0("w1[",i, "]"))
  samples_a11*w1
}

# a21*w1 = a21_star*sigma2*w1
cal_a21w1<- function(i){
  w1<-denmark_output_full$fit$draws(variables = paste0("w1[",i, "]"))
  samples_a21*w1
}

# a22*w2= sqrt(1-a21.star^2)*sigma2 * w2
cal_a22w2<- function(i){
  w2<-denmark_output_full$fit$draws(variables = paste0("w2[",i, "]"))
  samples_a22*w2
}

# eta1 = a11*w1
cal_eta1<- cal_a11w1

# eta2 = a21*w1+a22*w2
cal_eta2<- function(i){
  w1<- denmark_output_full$fit$draws(variables = paste0("w1[",i, "]"))
  w2<- denmark_output_full$fit$draws(variables = paste0("w2[",i, "]"))
  (samples_a21*w1) + (samples_a22*w2)
}


# w1
cal_w1<- function(i){
  denmark_output_full$fit$draws(variables = paste0("w1[",i, "]"))
}
samples_w1<- sapply(1:(grid_size^2), cal_w1)

# w2
cal_w2<- function(i){
  denmark_output_full$fit$draws(variables = paste0("w2[",i, "]"))
}
samples_w2<- sapply(1:(grid_size^2), cal_w2)


# samples for the decomposition
samples_a11_w1<- sapply(1:(grid_size^2), cal_a11w1)
samples_a21_w1<- sapply(1:(grid_size^2), cal_a21w1)
samples_a22_w2<- sapply(1:(grid_size^2), cal_a22w2)
samples_eta1<- samples_a11_w1
samples_eta2<- sapply(1:(grid_size^2), cal_eta2)


# add legend bar to each of the plot:

# --- helper for rotation ---
```

```r
rotate_reflect <- function(m) {
  t(apply(m, 2, rev))
}

# --- compute matrices ---
mat1 <- rotate_reflect(apply(matrix(apply(samples_beta0 + samples_a11_w1, 2, mean),
                                    nrow = grid_size, byrow = TRUE), 2, rev))
mat2 <- rotate_reflect(apply(matrix(apply(samples_a11_w1, 2, mean),
                                    nrow = grid_size, byrow = TRUE), 2, rev))
mat3 <- rotate_reflect(apply(matrix(apply(samples_beta1 + samples_eta2, 2, mean),
                                    nrow = grid_size, byrow = TRUE), 2, rev))
mat4 <- rotate_reflect(apply(matrix(apply(samples_a21_w1, 2, mean),
                                    nrow = grid_size, byrow = TRUE), 2, rev))
mat5 <- rotate_reflect(apply(matrix(apply(samples_a22_w2, 2, mean),
                                    nrow = grid_size, byrow = TRUE), 2, rev))

# --- define row-specific scales ---
vals_row1 <- c(mat1, mat2)
breaks_row1 <- seq(min(vals_row1), max(vals_row1), length.out = 16)
cols_row1   <- viridis(length(breaks_row1) - 1, option = "magma")

vals_row2 <- c(mat3,mat4, mat5)
breaks_row2 <- seq(min(vals_row2), max(vals_row2), length.out = 16)
cols_row2   <- viridis(length(breaks_row2) - 1, option = "magma")


# --- layout: 2 rows (3 plots + 2 plots) ---

layout(matrix(c(1,2,3,
                4,5,6), nrow=2, byrow=TRUE))

par(mar = c(4, 4, 2, 2))

# Row 1: for cell density
image.plot(mat1, col = cols_row1, breaks = breaks_row1,
           main = expression(log(lambda[1](g))),
           xlab = '', ylab = '', axes = FALSE,
           legend.mar = 4,
           legend.cex = 0.8)

image.plot(mat2, col = cols_row1, breaks = breaks_row1,
           main = expression(a[11]*w[1](g)),
           xlab = '', ylab = '', axes = FALSE,
           legend.mar = 4,
           legend.cex = 0.8)

plot.new()  # empty slot

# Row 2: for gene expression (shared scale)
image.plot(mat3, col = cols_row2, breaks = breaks_row2,
      main = expression(log(lambda[2](g))),
      xlab = '', ylab = '', axes = FALSE,
      legend.mar = 4,
```

```
      legend.cex = 0.8)

image.plot(mat4, col = cols_row2, breaks = breaks_row2,
           main = expression(a[21]*w[1](g)),
           xlab = '', ylab = '', axes = FALSE,
           legend.mar = 4,
           legend.cex = 0.8)

image.plot(mat5, col = cols_row2, breaks = breaks_row2,
           main = expression(a[22]*w[2](g)),
           xlab = '', ylab = '', axes = FALSE,
           legend.mar = 4,
           legend.cex = 0.8)
```

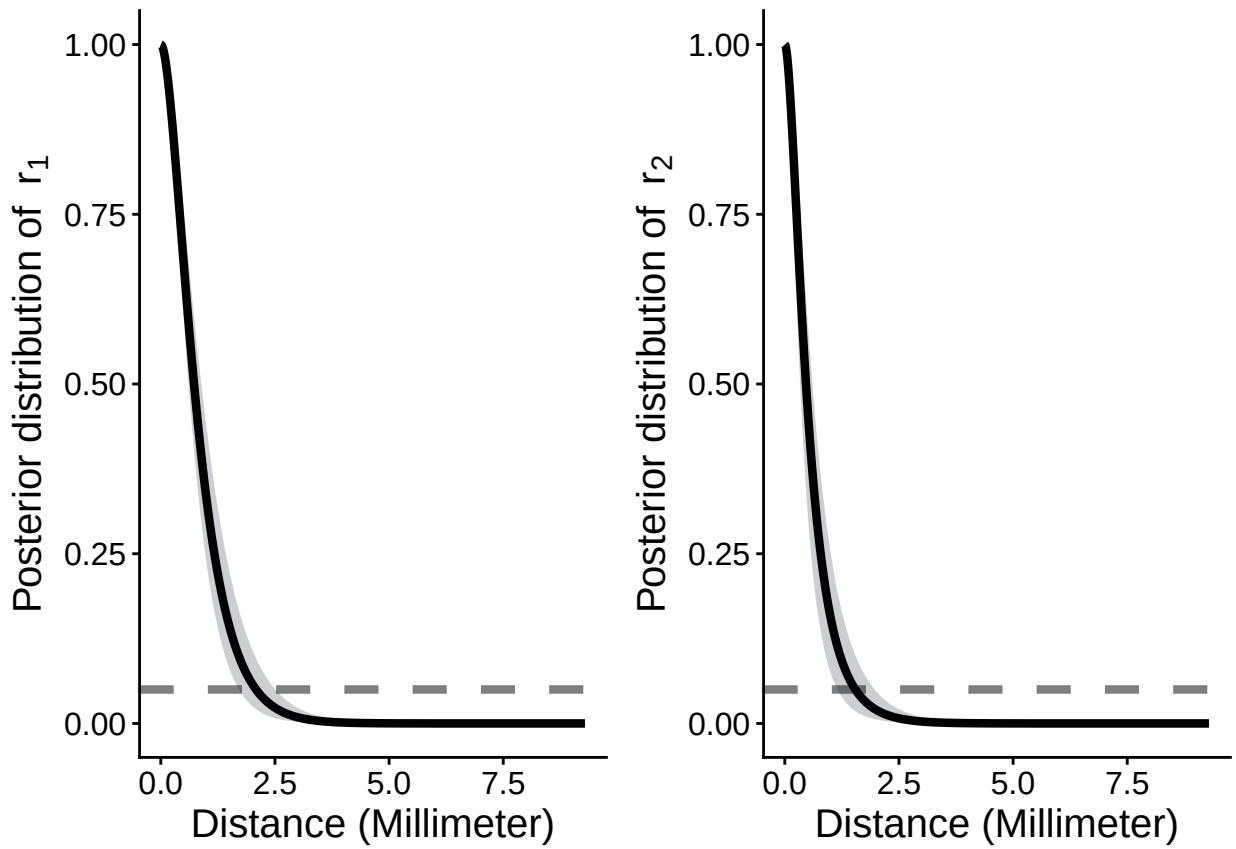


We might also want to check the spatial correlation functions for the two latent fields. This correlation measure how fast the two latent fields decay.

```
# [Code Block 3.11] Spatial correlation function

# a series of distance from (0, max(ROI distance)):
max_dist <- 2*sqrt(L1^2 + L2^2) # max distance of the ROI
dist_choice<- seq(0, max_dist, by=0.01)

# the samples from the DenMark full model
samples_rho1<- summary_full$`ell[1]`
samples_rho2<- summary_full$`ell[2]`
```

```r
samples_a21_HS<- summary_full$`sigma[2]`*summary_full$`Astar[2,1]`
samples_a22_HS<- summary_full$`sigma[2]`*sqrt(1- summary_full$`Astar[2,1]`^2)

# save the samples
post_rho1<- matrix(NA, nrow=length(dist_choice), ncol=length(samples_rho1))
post_rho2<- matrix(NA, nrow=length(dist_choice), ncol=length(samples_rho2))
post_mean_rho1<- rep(NA, length(dist_choice))
post_upper_rho1<- rep(NA, length(dist_choice))
post_lower_rho1<- rep(NA, length(dist_choice))
post_mean_rho2<- rep(NA, length(dist_choice))
post_upper_rho2<- rep(NA, length(dist_choice))
post_lower_rho2<- rep(NA, length(dist_choice))

for(k in 1:length(dist_choice)){
  post_rho1[k,]<- matern32(d=dist_choice[k], sigma=1, rho= samples_rho1)
  post_rho2[k,]<- (samples_a21^2*matern32(d=dist_choice[k],
                                          sigma=1,
                                          rho= samples_rho1)+
                     samples_a22^2*matern32(d=dist_choice[k],
                                            sigma=1,
                                            rho=samples_rho2))/(samples_a21^2+
                                                                samples_a22^2)
}

for (k in 1:length(dist_choice)){
  post_mean_rho1[k]<-mean(post_rho1[k,])
  post_upper_rho1[k]<-quantile(post_rho1[k,], 0.975)
  post_lower_rho1[k]<-quantile(post_rho1[k,], 0.025)

  post_mean_rho2[k]<-mean(post_rho2[k,])
  post_upper_rho2[k]<-quantile(post_rho2[k,], 0.975)
  post_lower_rho2[k]<-quantile(post_rho2[k,], 0.025)
}


# The latent field for cell density
df_rho1<- cbind( dist_choice ,
                 post_mean_rho1,
                 post_upper_rho1,
                 post_lower_rho1)

colnames(df_rho1)<- c('dist_choice',
                      'post_mean_rho1',
                      'post_upper_rho1',
                      'post_lower_rho1')

est_range_1<-dist_choice[which(abs(df_rho1[,2]-0.05)==min(abs(df_rho1[,2]-0.05)))]

eta1_corr<-ggplot(data=df_rho1, aes(x=dist_choice))+
  geom_ribbon(aes(ymin=post_lower_rho1, ymax= post_upper_rho1), fill = "#cbcfd2")+
  geom_line(aes(y=post_mean_rho1), color='#000', linewidth=1.5)+
  geom_hline(yintercept=0.05, linetype='dashed', col='#000', linewidth=1.5, alpha=.5)+
  xlim(0, 2*sqrt(L1^2 + L2^2))+
```

```r
  labs(x='Distance (Millimeter)', y=expression( 'Posterior distribution of '~ r[1]) )+
  theme_classic()+
  theme(axis.text.x = element_text(angle = -0, hjust = 0))+
  theme_classic() +
  theme(
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    axis.title.x = element_text(size = 15),
    axis.title.y = element_text(size = 15),
    legend.position = "none"
  )


# The latent field for gen expression
df_rho2<- cbind( dist_choice,
                 post_mean_rho2,
                 post_upper_rho2,
                 post_lower_rho2)
colnames(df_rho2)<- c('dist_choice',
                      'post_mean_rho2',
                      'post_upper_rho2',
                      'post_lower_rho2')
est_range_2<-dist_choice[which(abs(df_rho2[,2]-0.05)==min(abs(df_rho2[,2]-0.05)))]

eta2_corr<- ggplot(data=df_rho2, aes(x=dist_choice))+
  geom_ribbon(aes(ymin=post_lower_rho2, ymax= post_upper_rho2), fill = "#cbcfd2")+
  geom_line(aes(y=post_mean_rho2), color='#000', linewidth=1.5)+
  xlim(0, 2*sqrt(L1^2 + L2^2))+
  geom_hline(yintercept=0.05, linetype='dashed', col='black', linewidth=1.5, alpha=.5)+
  labs(x='Distance (Millimeter)', y=expression( 'Posterior distribution of '~ r[2]))+
  theme_classic()+
  theme(axis.text.x = element_text(angle = -0, hjust = 0))+
  theme_classic() +
  theme(
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    axis.title.x = element_text(size = 15),
    axis.title.y = element_text(size = 15),
    legend.position = "none"
  )

# show the figure:
plot_grid(eta1_corr, eta2_corr)
```




```
# The estimated ranges:
print(paste0('The etimated range for cell density is: ',
             round(est_range_1, 2),' mm.' ))
```

```
## [1] "The etimated range for cell density is: 2.09 mm."
```

```
print(paste0('The estimated range for gene expression is: ',
             round(est_range_2, 2), ' mm.' ))
```

```
## [1] "The estimated range for gene expression is: 1.55 mm."
```

4. DCG Discovery: Identify the density correlated genes in the dataset

This section, we will take the large range analysis as an example to show how the DCG discovery process is undertaken. This can be finished by running each candidate gene as the framework and select the genes with

- PCG: positive density-expression as positive density-correlated genes,
- NCG: genes with a negative density-expression correlation

5. Short range analysis

We now run DenMark for the short range analysis. The whole process is the same as that of the large range analysis, except for a smaller ROI window.

Section 2a introduce a small ROI, capable for short range analysis,

```
print('Under construction')
```

```
## [1] "Under construction"
```

```
# [Code Block 2a.1] Decide the ROI of interest

# selected invasive tumor ROI by pathologist: [0.5, 1.5]*[2, 3]
#center_slice_cell_location_invasive <- cell_spatial[which(cell_spatial$x>=0.5 & cell_spatial$x<= 1.5 &

# corresponding gene expression at the ROI
#gene_invasive<- cell_gene[which(cell_spatial$x>=0.5 & cell_spatial$x<= 1.5 & cell_spatial$y>= 2 & cell_
```

```
print('Under construction')
```

```
## [1] "Under construction"
```

```
# [Code Block 2a.2] Quality control

# filter out the genes with (1) the lowest 5% , (2) zeros with more than 5% locations
#gene_invasive <- gene_invasive[,-1]


# Convert to matrix
#mat <- as.matrix(gene_invasive)

# (1) Compute variance per gene
#gene_var <- apply(mat, 2, var)

# Keep top 95% high-variance genes
#var_threshold <- quantile(gene_var, 0.05)
#genes_high_var <- names(gene_var[gene_var >= var_threshold])

#
# (2) Non-zero filter (>=5%)
#nonzero_fraction <- colSums(mat > 0) / nrow(mat)
#genes_nonzero <- names(nonzero_fraction[nonzero_fraction >= 0.05])

#
# (3) Combine both filters
#genes_keep <- intersect(genes_high_var, genes_nonzero)

# (4) Filtered dataset
#gene_invasive_filtered <- mat[, genes_keep, drop = FALSE]

# check the filtered genes
#colnames(gene_invasive_filtered)[1:10]
```